

Lecture 16

Saturday, April 11, 2026 9:21 AM

Today: Logistic Regression Classification

Given a data point $\rightarrow$ output label

Examples:

- Spam filter: Is the email spam ($y=1$) or not ($y=0$)
- Medical diagnosis: Does the patient have this condition or not
- Fraud detection: Is this transaction fraudulent or not.

Training data

$$(x^{(i)}, y^{(i)}) \quad 1 \leq i \leq N$$

$$y^{(i)} \in \{0, 1\}$$

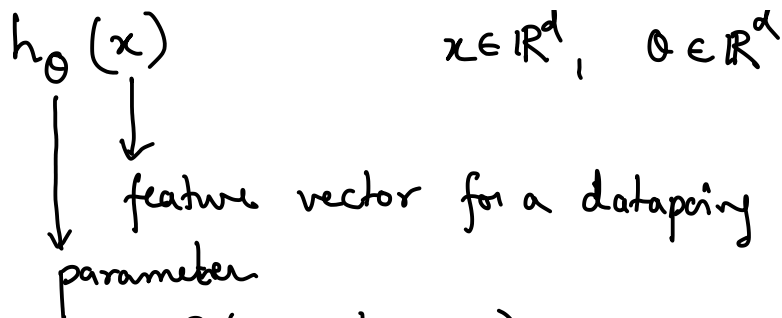
$$x^{(i)} \in \mathbb{R}^d \quad (d \text{ features})$$

Examples:

- ① $x^{(i)}$: systolic blood pressure reading
 $y^{(i)}$: chronic high blood pressure present?
- ② "buy" "click" "free" "unsubscribe" $d=10000$
 frequency of the i^{th} word in the email
 $y^{(i)} = \begin{cases} 1 & \text{spam} \\ 0 & \text{not spam} \end{cases}$

Logistic regression:

Solves this problem using a hypothesis function.



$h_{\theta}(x)$ measures $P(y=1 | x; \theta)$
 $\downarrow$ $\downarrow$
data point parameter

$$h_{\theta}(x) : \mathbb{R}^d \rightarrow [0, 1]$$

① $0 \leq h_{\theta}(x) \leq 1$ $(h_{\theta}(x) = p(y=1 | x; \theta))$

② x 's score : $\theta^T x$ ($\in \mathbb{R}$)

a larger value of $\theta^T x \Rightarrow h_{\theta}(x) \rightarrow 1$

a smaller value of $\theta^T x \Rightarrow h_{\theta}(x) \rightarrow 0$

③ $h_{\theta}(x) = 1/2$ decision threshold

$h_{\theta}(x) < 1/2 \quad \hat{y} = 0$

$h_{\theta}(x) > 1/2 \quad \hat{y} = 1$

What is form of $h_{\theta}(x)$?

Candidate 1: $h_{\theta}(x) = \theta^T x$ $\leftarrow$ does not satisfy property ①

$$0 \leq h_{\theta}(x) \leq 1$$

Instead of probability, look at odds

The odds of this stock going up are 3-to-1
 $\Leftrightarrow$ " probability " " " " " " is 75% or 0.75

$$\text{odds} = \frac{h_{\theta}(x)}{1 - h_{\theta}(x)} \in (0, \infty)$$

Candidate 2: $\frac{h_{\theta}(x)}{1 - h_{\theta}(x)} = \theta^T x$ $\leftarrow$ does not satisfy property ①

Take logs

$$\log \text{ odds} = \log \left(\frac{h_{\theta}(x)}{1 - h_{\theta}(x)} \right) \in (-\infty, \infty)$$

Logistic regression assume:

$$\log \left(\frac{h_{\theta}(x)}{1 - h_{\theta}(x)} \right) = \theta^T x \quad \text{--- (1)}$$

$\downarrow$ parameters $\rightarrow$ data

Event E

$P(E)$: probability of event E

odd (E): $\frac{P(E)}{1 - P(E)}$

From (1): $\frac{h_{\theta}(x)}{1 - h_{\theta}(x)} = \exp(\theta^T x)$

$$\Rightarrow h_{\theta}(x) [1 + \exp(\theta^T x)] = \exp(\theta^T x)$$

$$\Rightarrow h_{\theta}(x) = \frac{\exp(\theta^T x)}{1 + \exp(\theta^T x)} \quad \text{--- (2)}$$

$$P(y=1|x; \theta) = h_{\theta}(x) = \frac{1}{1 + \exp(-\theta^T x)} = g(\theta^T x)$$

$$g(z) = \frac{1}{1 + e^{-z}} \quad \left. \vphantom{g(z)} \right\} \text{sigmoid function}$$

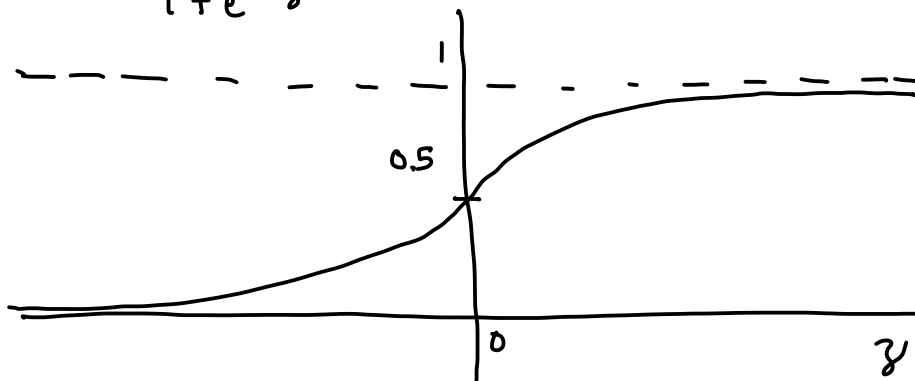
tl; dr

$$P(y=1|x; \theta) = h_{\theta}(x) = \frac{1}{1 + \exp(-\theta^T x)}$$

Modeling assumption of logistic regression

Properties of the sigmoid function

$$g(z) = \frac{1}{1 + e^{-z}}$$



$$\left. \begin{array}{l} \textcircled{1} g'(z) = g(z)(1-g(z)) \\ \textcircled{2} 1-g(z) = g(-z) \end{array} \right\} \text{ will be important later}$$

Summary:

Logistic regression assumes

$$h_{\theta}(x) = g(\theta^T x) = \frac{1}{1 + e^{-\theta^T x}} = P(y=1 | x; \theta)$$

↓
Sigmoid function

Q1: How to decide the value of θ ?

Q2: How to compute $\hat{y}$ from $h_{\theta}(x)$?
 $\hat{y} \in \{0, 1\}$

Ans 2: θ is known

a new sample x comes, $\hat{y} = ?$

$$h_{\theta}(x) = \begin{cases} > 1/2 & \hat{y} = 1 \\ < 1/2 & \hat{y} = 0 \\ = 1/2 & \text{toss a fair coin} \end{cases} \quad \left. \vphantom{h_{\theta}(x)} \right\} \text{Rule 1}$$

$$h_{\theta}(x) = \frac{1}{1 + \exp(-\theta^T x)} = P(y=1 | x; \theta)$$

$$h_{\theta}(x) > \frac{1}{2} \Rightarrow \theta^T x > 0$$

$$\left(\frac{1}{2} < \frac{1}{1 + \exp(-\theta^T x)} \right)$$

$$\Rightarrow 1 + \exp(-\theta^T x) < 2$$

$$\Rightarrow \exp(-\theta^T x) < 1$$

$$\Rightarrow -\theta^T x < 0$$

$$\Rightarrow \theta^T x > 0$$

$$h_{\theta}(x) < \frac{1}{2} \Rightarrow \theta^T x < 0$$

Rule 1 $\equiv$ Rule 2 : Predict $\hat{y}=1$ if $\theta^T x > 0$
 Predict $\hat{y}=0$ if $\theta^T x < 0$
 Toss a coin if $\theta^T x = 0$

Back to Q1: How to compute θ ?

Assumptions:

- ① Training data samples are i.i.d.
- ② $P(y=1|x; \theta) = h_\theta(x) = g(\theta^T x) = \frac{1}{1 + \exp(-\theta^T x)}$
- ③ Classes are Bernoulli random variables

$$y|x; \theta \sim \text{Ber}(h_\theta(x))$$

$$\Rightarrow y|x; \theta = \begin{cases} 1 & \text{w.p. } h_\theta(x) \\ 0 & \text{w.p. } 1-h_\theta(x) \end{cases}$$

Likelihood function

$L(\theta)$: measure of what is the probability of seeing our training data given θ .

$$\rightarrow L(\theta) = \prod_{i=1}^N P(y^{(i)} | x^{(i)}; \theta) \quad \text{--- (3)}$$

$$\rightarrow \theta^* = \underset{\theta}{\operatorname{argmax}} L(\theta) \quad \leftarrow \text{a principled way to obtain } \theta^*$$

$$P(y^{(i)} = 1 | x^{(i)}; \theta) = h_\theta(x^{(i)}) \quad \text{--- (4)}$$

$$P(y^{(i)} = 0 | x^{(i)}; \theta) = 1 - h_\theta(x^{(i)}) \quad \text{--- (5)}$$

$$\Rightarrow P(y^{(i)} | x^{(i)}; \theta) = h_\theta(x^{(i)})^{y^{(i)}} (1 - h_\theta(x^{(i)}))^{(1-y^{(i)})} \quad \text{--- (6)}$$

$$\text{(4) \& (5) } \Leftrightarrow \text{(6) . Using (3) \& (6)}$$

$$L(\theta) = \prod_{i=1}^N h_{\theta}(x^{(i)}) y^{(i)} (1 - h_{\theta}(x^{(i)}))^{(1-y^{(i)})} \quad (7)$$

$$\theta^* = \arg \max_{\theta} L(\theta)$$

Some manipulations to make our life simple, log of (7)

$$\log L(\theta) = \sum_{i=1}^N y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)}))$$

$$\Rightarrow \begin{array}{l} \text{maximize } \log L(\theta) \\ \text{minimizing } -\log L(\theta) = J(\theta) \end{array}$$

$$J(\theta) = - \sum_{i=1}^N y^{(i)} \log h_{\theta}(x^{(i)}) + (1 - y^{(i)}) \log (1 - h_{\theta}(x^{(i)}))$$

① $J(\theta)$ is convex

② $J(\theta)$ is called the cross-entropy function

$$\theta^* = \arg \min_{\theta} J(\theta)$$

Gradient descent or any of its variants can be used.

$$\theta \leftarrow \theta - \frac{\alpha}{N} x^T (g(x^T \theta) - y)$$

